

PROGRAMMING IN OPHTHALMOLOGY · SESSION 2

Reading Models Critically and Evaluating Results

Recap · what we did not finish in Session 1 · confusion matrix, ROC, cross-validation, confidence intervals

Dr. Simon König · Online Workshop, 12 November 2026, 18:00 GMT

Teaching material - not a validated clinical tool

0 WELCOME

Today: recap, what we missed, and how to trust a result



0:45 - 1:00 Questions & discussion (evaluation questions, your own data)

How we will work

- Slides explain a concept - then we switch to the notebook and run it
- Everything is in one GitHub repository: slides (PDF) and notebooks, with Colab / Kaggle / local instructions
- No programming needed: you are the clinical expert, the code is a tool

POLL · answer in the chat

Which topic of Session 1 should we repeat?

A) what is ONE row / merging
B) splitting by patient
C) scaling / models

R

RECAP · SESSION 1

Where we stopped: from raw tables to a first model

1

ONE row

What is the unit of analysis? Rows vs patients vs eyes

2

Merge & clean

Merge by key, validate, log every cleaning decision

3

Split by patient

Never put the same patient in training and test

4

Scale on train only

Fit preprocessing on training data, apply to test

5

Interpretable model

Logistic regression, small tree - read them critically

The one question behind all five steps

What was independent between training and test - and was it enough? Today we add: how do we measure the result, and how sure are we?

R

RECAP · QUICK QUIZ

Three questions - answer in the chat

Q1

A table has 2,000 eye examinations from 700 patients. A random split of ROWS gives 97 % test accuracy. What is wrong?

A) nothing B) the same patient can be in training and test C) 97 % is too low

Q2

When do you fit the StandardScaler (mean and SD)?

A) on all data, before splitting B) on the training set only C) separately on test

Q3

One feature alone separates the groups almost perfectly (AUC 0.99). What do you ask first?

A) which model is best? B) how was the label defined? C) can I add more features?

B

FROM SESSION 1 · MODULE 2

Reading a model's reasoning critically

Sign conventions

Octopus: positive MD = loss. Humphrey: negative MD = loss. A 'positive coefficient' means opposite things in different datasets.

Correlated features

RNFL and MRW carry similar information. Add or remove one and the other's coefficient jumps - interpret groups of features, not single numbers.

Circular labels

If one feature almost perfectly separates the groups, ask how the label was defined. A label built from the features teaches the model nothing new.

Association ≠ causation

'Important to the model' is not 'important to the eye'. A model sees correlations in THIS dataset only.

SWITCH TO NOTEBOOK · B6

B

FROM SESSION 1 · MODULE 2

Checklist: conceiving your own ML analysis

1

Question

Which clinical decision would this support?

2

Unit of analysis

What is ONE row? What must stay independent?

3

Label

Who defined it, when, how - independent of the features?

4

Features

Available at the moment of use? Nothing from the future?

5

Split

By patient (or site, or time) - never by row

6

Metric & baseline

Sensitivity/specificity vs. 'always predict the majority'

7

Failure analysis

Subgroups, correlated features, label circularity

8

Before clinical use

External validation, prospective test, regulation

SWITCH TO NOTEBOOK · B7

SPOT THE FLAW · 0:14 - 0:19

Spot the flaw - three short cases

Case 1

A group trains a model on 12,000 OCT B-scans from 900 patients. They randomly assign 75 % of the IMAGES to training and 25 % to testing and report 98 % accuracy.

Case 2

A model predicts 'glaucoma' from RNFL thickness and visual-field MD with 97 % accuracy. Methods: "glaucoma was defined as RNFL below the 5th percentile and MD worse than -2 dB."

Case 3

To make the data comparable, all values are standardised using the mean and SD of the WHOLE dataset. Afterwards the data are split into training and test sets.

What is wrong - and how would you fix the design?

SPOT THE FLAW · 0:18 - 0:20

Spot the flaw - possible answers

Case 1 · Patient-level leakage

Images of the same patient (and both eyes) sit in both sets. Split by PATIENT and report that result. (Session 1, notebook B2.)

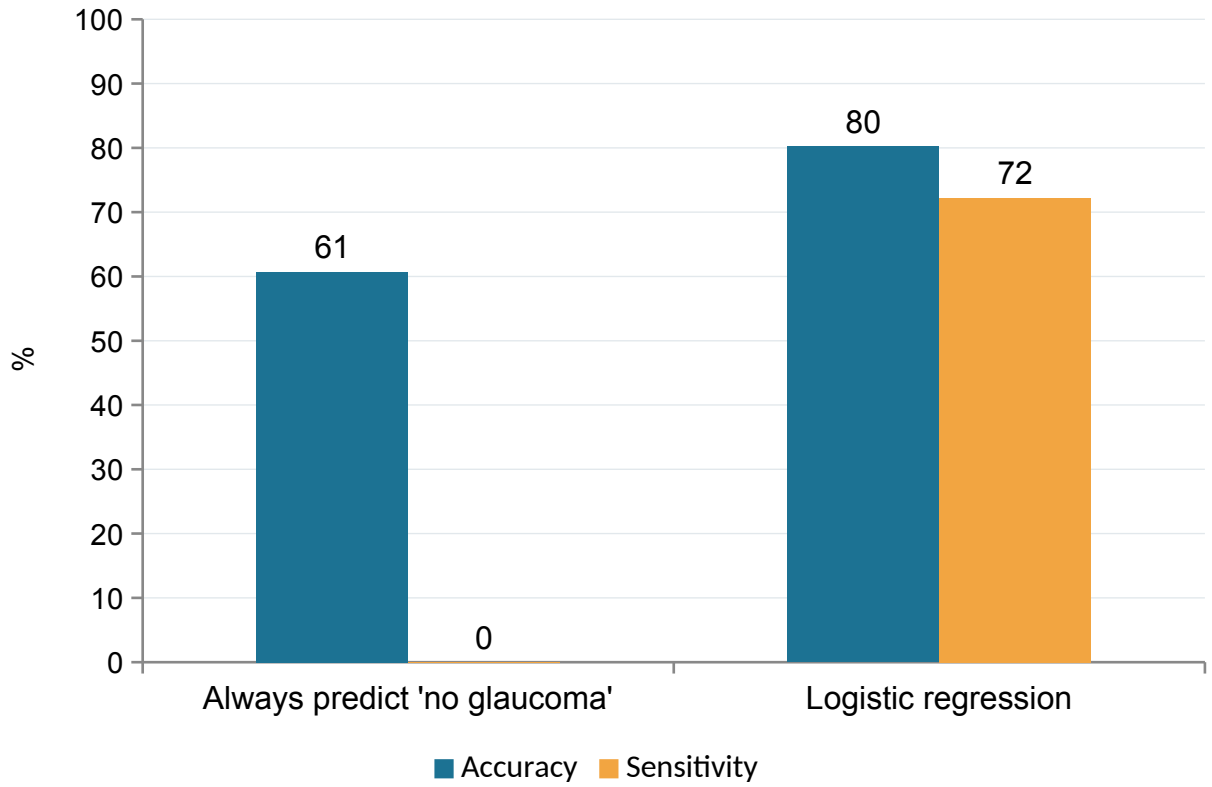
Case 2 · Circular label

The label is defined by the features, so the model only re-learns the rule. Use an independent reference standard (specialists blinded to the measurements) - or change the question, e.g. predict progression. (Notebook B6.)

Case 3 · Preprocessing leakage

Test-set information entered the scaling. Fit scaling - and any imputation or feature selection - on the TRAINING set only, ideally inside one pipeline. (Session 1, notebook B4.)

Accuracy alone can hide the failure



Public notebook, test patients. Model sees age, sex and one visual-field point.

A 'model' that does nothing scores 61 %

If 61 % of eyes are healthy, always answering 'healthy' is 61 % accurate - and misses every glaucoma eye (sensitivity 0 %).

Always report

- Sensitivity and specificity, not accuracy alone
- A simple baseline to beat
- Results on patients the model has never seen

The confusion matrix: four outcomes, many metrics

	predicted: glaucoma	predicted: healthy
truly glaucoma	TP **213** caught	FN **82** missed case
truly healthy	FP **67** false alarm	TN **389** correctly cleared

Public notebook, 751 test rows (unseen patients), threshold 0.5.

Sensitivity — `TP / (TP + FN)` — **0.72**
How many glaucoma eyes do we catch?

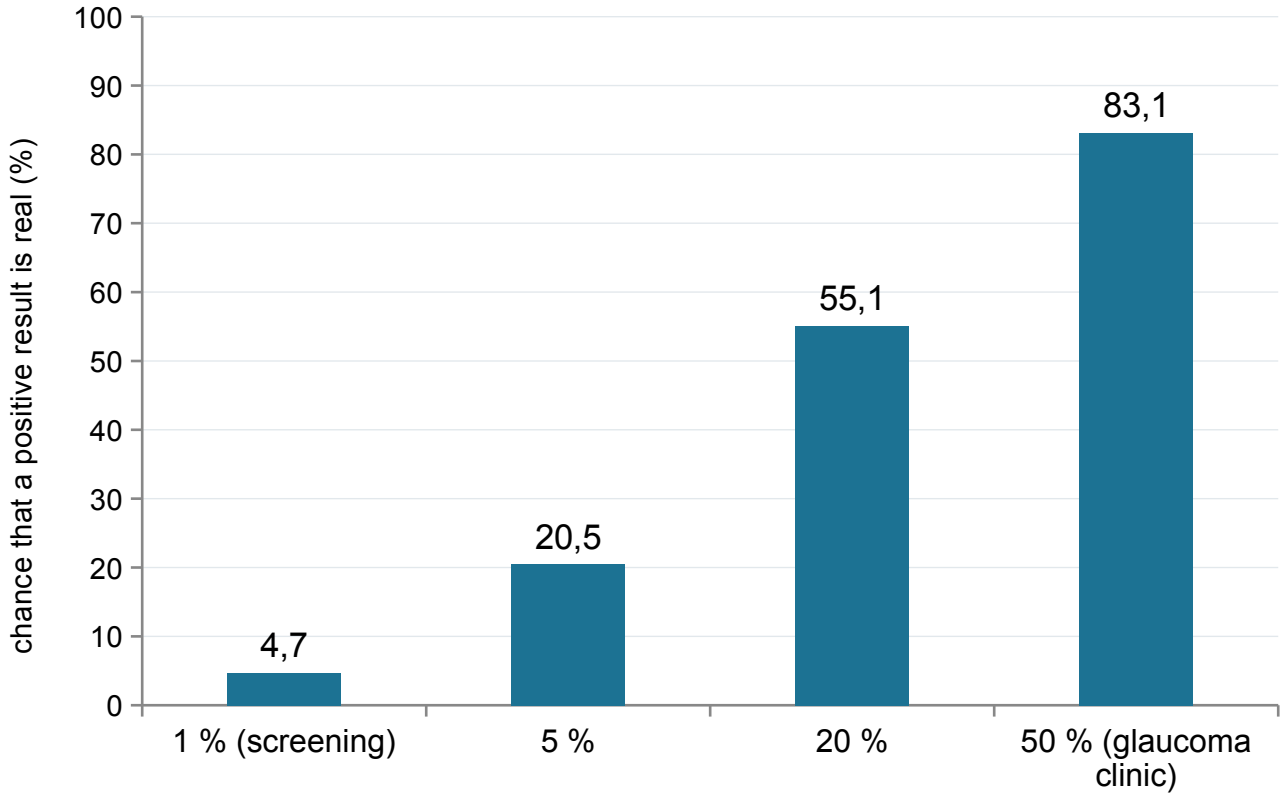
Specificity — `TN / (TN + FP)` — **0.85**
How many healthy eyes do we clear?

PPV (precision) — `TP / (TP + FP)` — **0.76**
If the model says 'glaucoma', how often is it right?

Accuracy — `(TP + TN) / all` — **0.80**
Overall hit rate - hides the error types

SWITCH TO NOTEBOOK · D1

Same test, different meaning: prevalence changes PPV



Sensitivity 0.72 and specificity 0.85 stay the same - only the prevalence changes. Public notebook model.

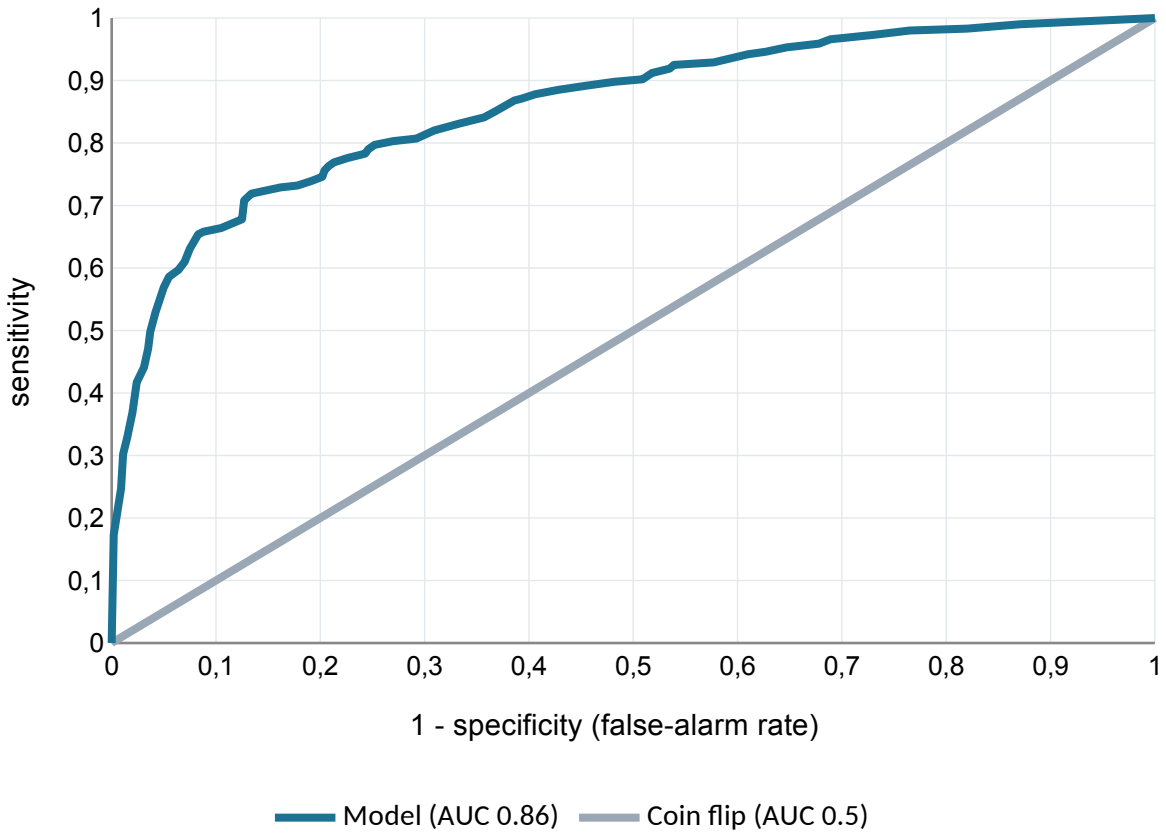
A positive screening result is often a false alarm

At 1 % prevalence, about 95 of 100 positive results are false positives - even with a decent test.

Ask of every paper

In which population was this measured - and is it the population where the model will be used?

ROC curve and AUC: all thresholds at once



AUC: 0.5 = coin flip, 1.0 = perfect. The threshold is a clinical decision, not a statistical one.

0.2

Low threshold

sensitivity 0.90, specificity 0.49 - catches almost everything, many false alarms: screening

0.5

Default threshold

sensitivity 0.72, specificity 0.85 - a compromise nobody chose on purpose

0.8

High threshold

sensitivity 0.48, specificity 0.96 - few false alarms, many misses: confirmatory test

SWITCH TO NOTEBOOK · D2

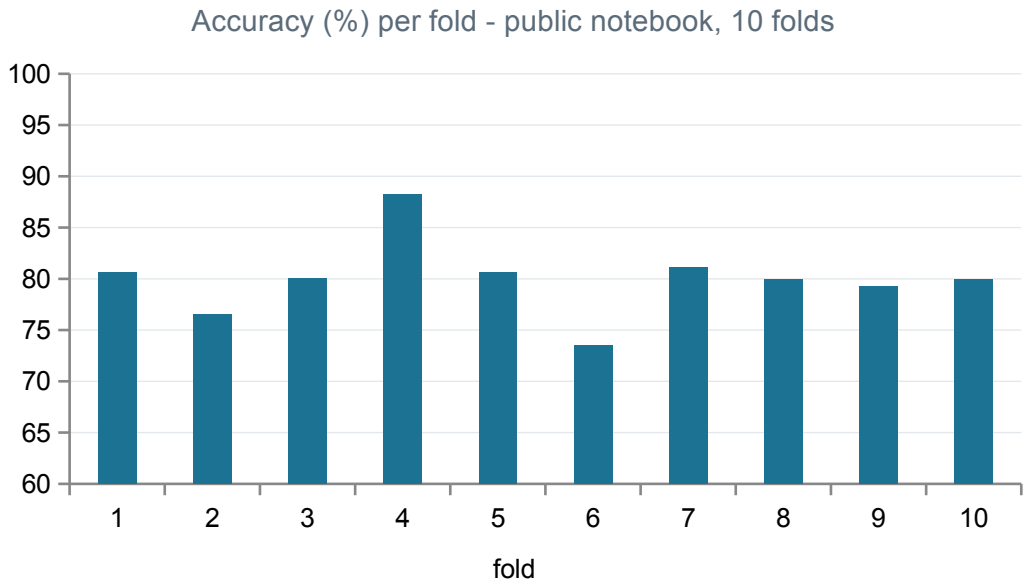
One split is one draw: cross-validation

Each patient is in the test fold exactly once (shown: 5 folds; the notebook uses 10)

Fold 1	**test**	train	train	train	train
Fold 2	train	**test**	train	train	train
Fold 3	train	train	**test**	train	train
Fold 4	train	train	train	**test**	train
Fold 5	train	train	train	train	**test**

A block = a group of PATIENTS (all rows of a patient stay together)

Scaling and other preprocessing must be re-fitted inside every training fold - use a pipeline.



mean 80.0 %, SD 3.7 %, range 73.5 - 88.2 % - the result depends on who is in the test set

SWITCH TO NOTEBOOK · D3

How sure are we? Bootstrapped confidence intervals

1 Draw patients

from the test set, WITH replacement - same size as the original

2 Recompute

accuracy, AUC, sensitivity… on this resample

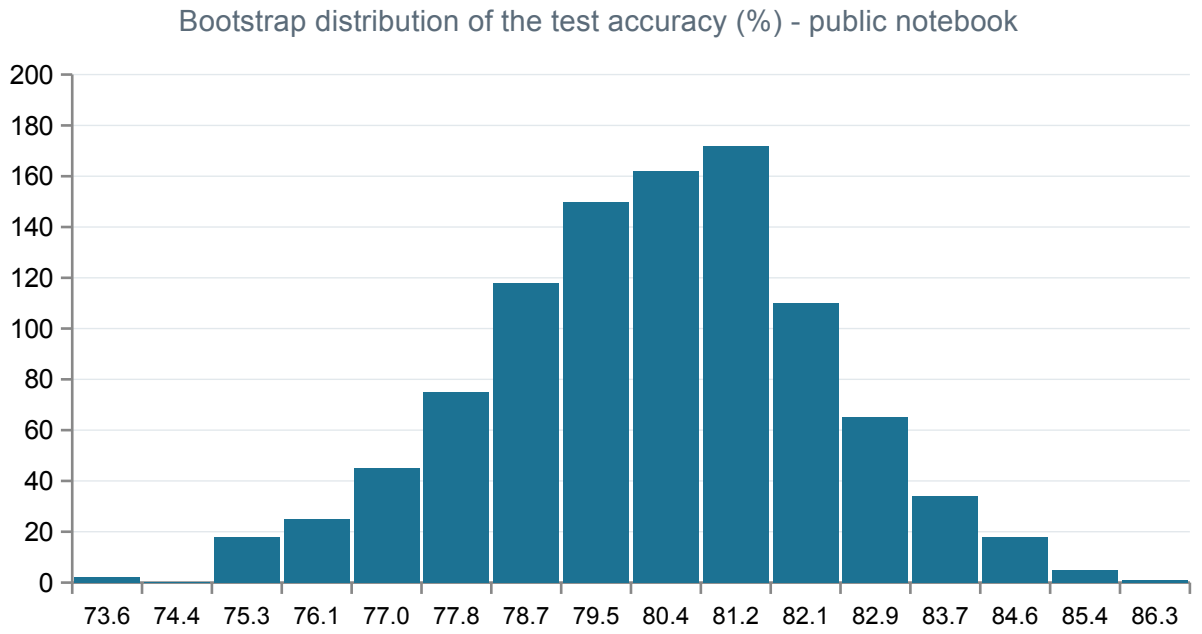
3 Repeat 1,000x

and look at the spread of the results

4 Read the 95 % CI

2.5th to 97.5th percentile

Resample PATIENTS, not rows - otherwise the interval is too narrow.



Accuracy 80.2 % (95 % CI 75.9 - 84.1 %)

Sensitivity 72.2 % (63.9 - 79.9 %) · specificity 85.3 % (81.2 - 89.1 %)

SWITCH TO NOTEBOOK · D4

How to report a result

✗ Not convincing

"Our model detects glaucoma with 98 % accuracy."

No split described, no baseline, no sensitivity/specificity, no uncertainty, no population.

✓ Reads like a result

"On 100 unseen patients (patient-wise split), sensitivity was 72 % (95 % CI 64-80 %) and specificity 85 % (81-89 %); baseline accuracy 61 %."

(public notebook example)

Metric

sensitivity, specificity, PPV - and AUC

Uncertainty

confidence interval from patients

Design

how the split was made, what was independent

Baseline

what a trivial rule would score

WRAP-UP

Take-home messages

1. Read coefficients critically: sign convention, correlated features, circular labels
2. Report sensitivity and specificity (and PPV) - and always a baseline
3. The threshold is a clinical decision; the ROC curve shows all trade-offs
4. Use patient-wise cross-validation - one split is one draw
5. Add a confidence interval from resampled patients - small test sets mean wide intervals

Next session

Deep learning and computer vision

- Regression metrics (MAE, RMSE, R^2)
- Neural networks: how they learn
- Images and convolutional networks (OCT)
- Heatmaps: what did the network look at?

?

0:45 - 1:00

Questions & discussion (15 min)

1 Which error is worse?

A missed glaucoma eye or a false alarm - in your clinic, with your referral options?

2 Which threshold would you choose?

For a screening campaign vs. for a confirmatory test.

3 Would you trust this paper?

No confidence interval, no baseline, split not described - what would you ask the authors?

4 Which data do you have?

Could you test a model in your population - and who would have to agree to share the data?

Type questions in the chat · or raise your hand

AFTER THE SESSION

Everything is in the GitHub repository

1

Open the repository

github.com/simonkng/ophthalmology-ml-workshop

2

Open the notebook with the 'Open in Colab' badge

Google account needed - runs in the browser, nothing to install

3

Or use Kaggle / run locally

Step-by-step instructions in SETUP.md; one-page cheat sheet in /cheatsheet

4

Runtime → Run all

The first run downloads a small public CSV (internet needed)

Scan to open the repository

Data & references

- Public notebooks: Harvard-GDP (CC BY-NC-ND 4.0), downloaded from its source - nothing redistributed
- Luo Y, Shi M, Tian Y, Elze T, Wang M. Harvard Glaucoma Detection and Progression. ICCV 2023
- Follow-up visits in the recap are simulated
- Libraries: pandas, scikit-learn, matplotlib, seaborn

Questions after the session: simon.koenig@unimedizin-mainz.de · One-page cheat sheet: see the repository